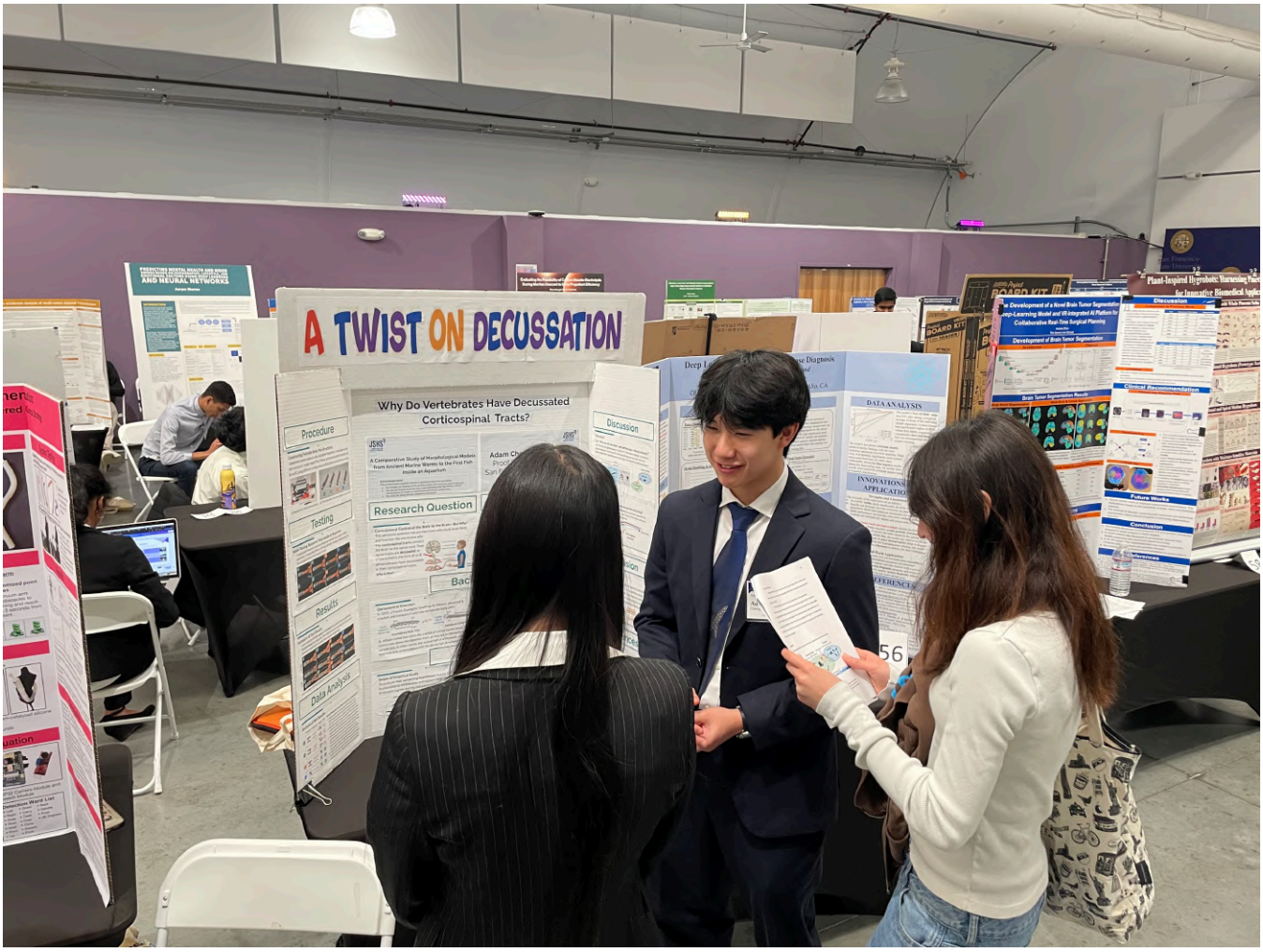
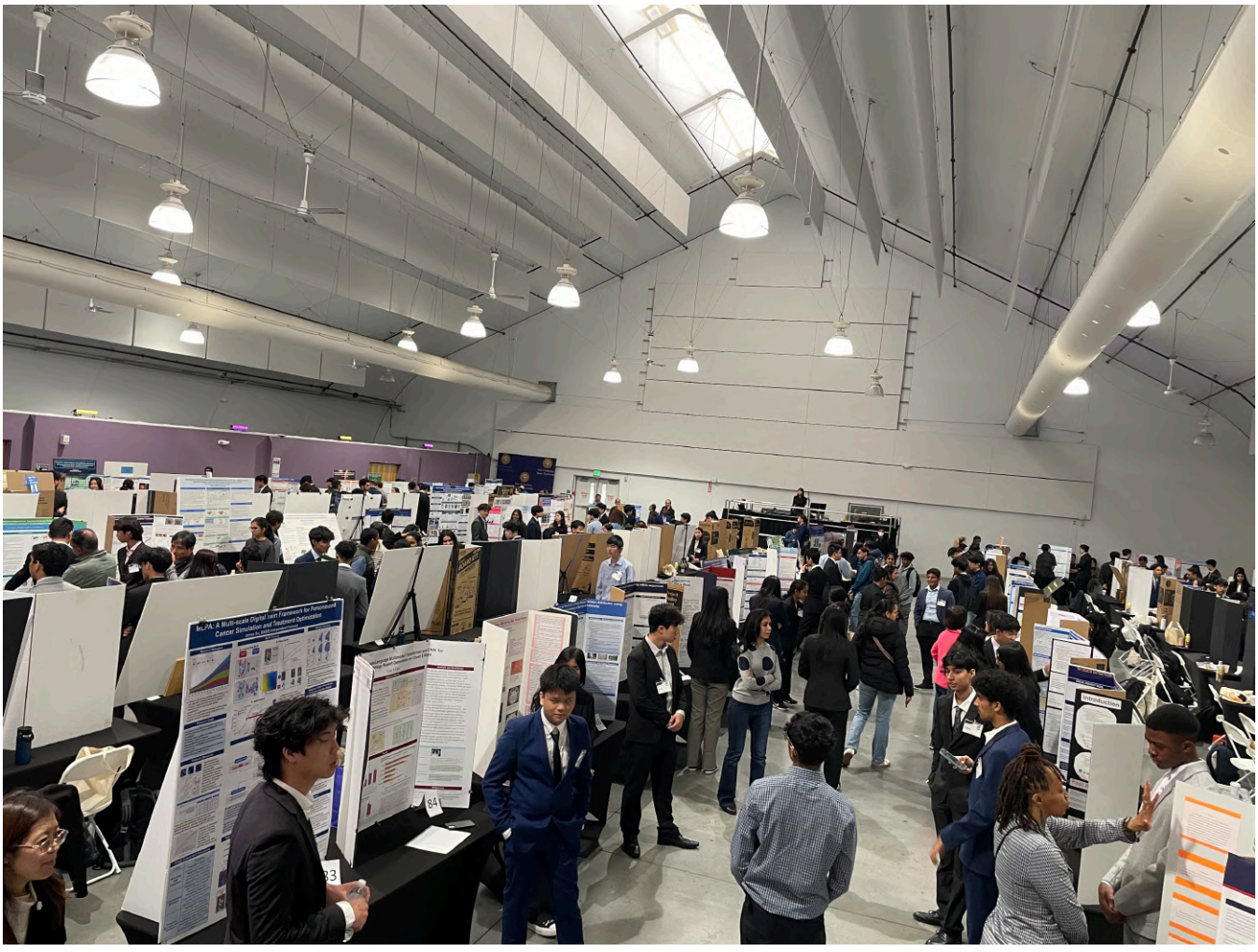

A TWIST ON DECUSSATION
Why Do Vertebrates Have Decussated Corticospinal Tracts?
Procedure
Testing
Results
Data Analysis
Research Question
Discussion



“Why Do Vertebrates Have Decussated Corticospinal Tracts?” — A Comparative Study of Morphological Models From Ancient Marine Worms to the First Fish Inside an Aquarium

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What if we could recreate evolutionary history inside an aquarium? What would that tell us about the greatest mystery in evolutionary biology: why do vertebrates have decussation in their corticospinal tracts?

The reason why our nerve tracts are *decussated* is buried deep in our evolutionary past. I investigated two leading theories: axial twist theory and somatic twist theory, that describe how the vertebrate body plan arose from the invertebrates 550 million years ago.

To test the two competing hypotheses, I built morphological models for empirical study within an aquarium environment. I chose the annelid as a reference model for ancient marine worms, who last shared a common ancestor with the lancelet, a closest living proxy of *Pikaia*. I constructed three ancient marine worm models with distinct topological configuration of the oropharynx and nerve cords inside a twistable body plan structure, alongside a Pikaia model.

I conducted buoyancy tests by dropping each model into the aquarium. The effects of body density changes underwater were noted. When the stomach goes from empty to full, I observed decussation resulting from dorsoventral inversion as predicted by somatic twist theory. However, all attempts to reproduce axial twists had failed.

My study reveals how the gullet escaped the invertebrate nerve ring, through a *transitory* body plan adapting to bountiful food sources, followed by a *somatic twist* leading to decussation. My investigation uncovers an evolutionary role for *deuterostomy* that leads to decussated nerve tracts in vertebrates, and opens up new pathways for future research in evolutionary biology.